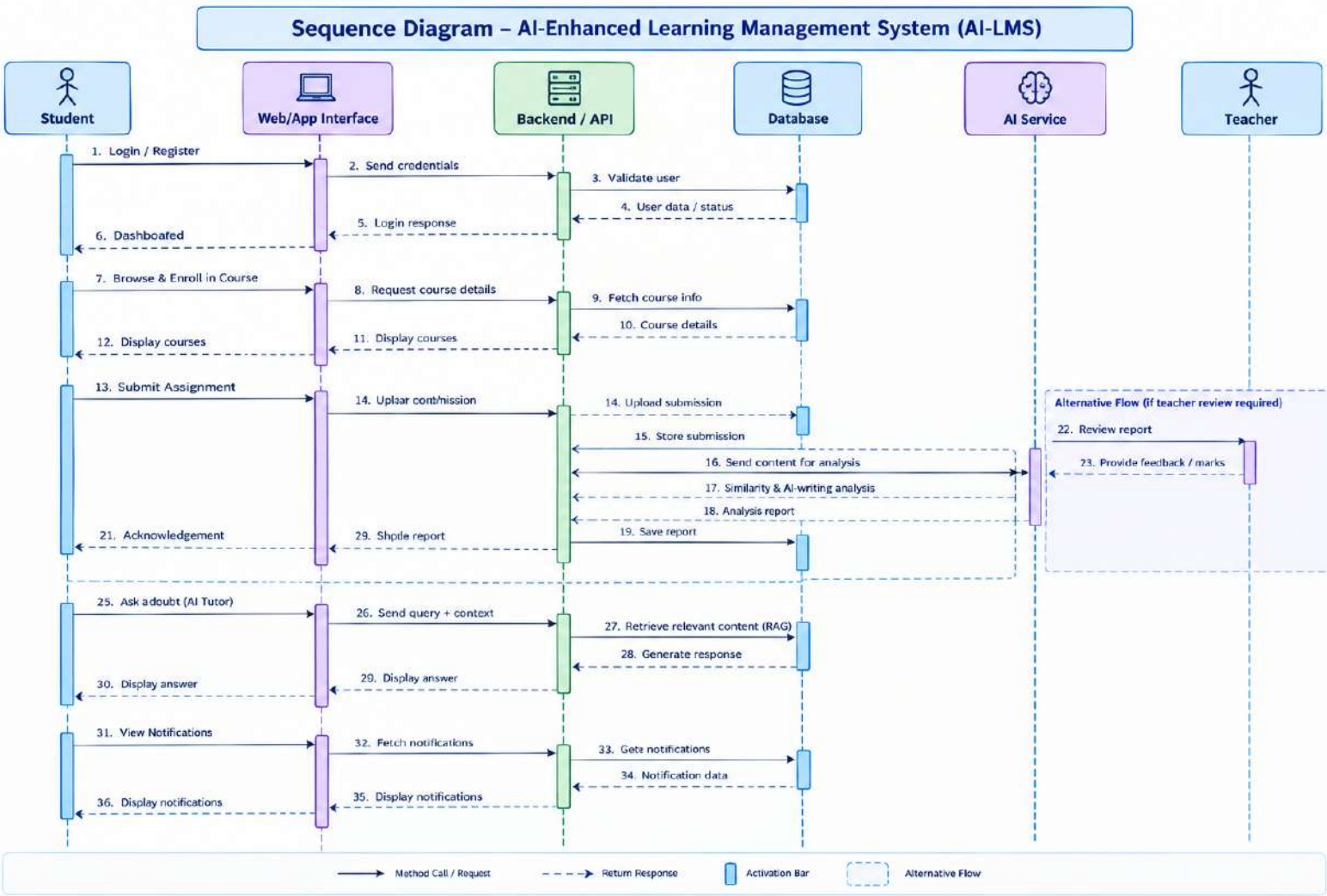

Sequence Diagram – AI-Enhanced Learning Management System (AI-LMS)
Student
Web/App Interface
Backend / API
Database
AI Service
Teacher
1. Login / Register
2. Send credentials
3. Validate user
4. User data / status
5. Login response
6. Dashboated
7. Browse & Enroll in Course
8. Request course details
9. Fetch course info
10. Course details
11. Display courses
12. Display courses
13. Submit Assignment
14. Upload submission
15. Store submission
16. Send content for analysis
17. Similarity & AI-writing analysis
18. Analysis report
19. Save report
21. Acknowledgement
29. Shpde report
Alternative Flow (if teacher review required)
22. Review report
23. Provide feedback / marks
25. Ask a doubt (AI Tutor)
26. Send query + context
27. Retrieve relevant content (RAG)
28. Generate response
29. Display answer
30. Display answer
31. View Notifications
32. Fetch notifications
33. Gete notifications
34. Notification data
35. Display notifications
36. Display notifications
Method Call / Request
Return Response
Activation Bar
Alternative Flow